

# ASSESSING KNOWLEDGE, ADOPTION WILLINGNESS, AND SKILL ACQUISITION PREFERENCES FOR GENERATIVE AI AMONG EMPLOYEES IN SMALL TO MEDIUM-SIZED ENTERPRISES

Ionuț-Alberto Boțoroga, Emil Fox, Endijs Kiršteins and Carlijn van der Vleuten

## 1 INTRODUCTION

As artificial intelligence (AI), particularly generative AI, continues to transform industries, businesses are seeking ways to integrate AI into their operations to enhance productivity and remain competitive. For small to medium-sized enterprises (SMEs), the challenge is not only adopting generative AI, but also ensuring that their workforce is adequately equipped to leverage these technologies effectively.

Employees in these organizations may have varying degrees of familiarity with generative AI, from little to no experience to some prior exposure. The roles of these employees could benefit from a basic understanding of using generative AI tools, such as text generation, image generation and other technologies. Given the constraints such as limited resources, time, and access to formal training, it is essential for SMEs to identify effective methods for teaching AI to employees.

This study aims to identify effective methods for teaching generative AI to employees. By understanding the employees' current knowledge levels, willingness to learn, and preferred learning methods, SMEs can create effective strategies to teach their employees and capitalize on the potential of generative AI.

### 1.1 Research Question

*What is the current level of knowledge about generative AI among employees in small to medium-sized companies, how willing are they to adopt it in their roles, and what is their preferred way of acquiring these skills?*

To provide an answer to this research question, we have come up with the following sub-questions:

1. How do negative concerns about Data Sciences and Artificial Intelligence impact SME employees' willingness to learn about Data Sciences and Artificial Intelligence.
2. What is the relationship between the willingness among employees in small to medium-sized companies to adopt generative AI in their roles and their age?
3. What are the main concerns or misconceptions employees have about AI?
4. How is an SME employee's existing knowledge of generative AI associated with their willingness to pursue further learning about the usage of generative AI?

### 1.2 Problem Statement

The adoption of generative AI in small to medium-sized enterprises is affected by a lack of understanding about the current level of knowledge and willingness of employees to adopt this technology in their roles as well as their preferred methods for acquiring the necessary skills. This knowledge gap affects not only the employees themselves, who are concerned

about job security and the relevance of their skills, but also SME owners and managers, who are responsible for making strategic decisions about AI adoption and are concerned about return on investment, employee retention, and competitive advantage.

Furthermore, the successful integration of generative AI in SMEs also depends on the support of industry associations and chambers of commerce, government and policymakers, educational institutions and training providers, AI and technology developers, and customers and clients of SMEs. These stakeholders have varying levels of power, attention, and interest in generative AI adoption, but all play a critical role in shaping the ecosystem that enables or hampers the adoption of generative AI in SMEs.

By investigating the current level of knowledge, willingness, and preferred learning methods among employees in SMEs, this research aims to provide valuable insights that can inform strategies for successful adoption and integration of generative AI in these organizations, ultimately benefiting all stakeholders involved.

## 2 LITERATURE STUDY

Research shows that when adopting AI into SMEs, things like employee attitudes, organizational culture, and the needed resources all influence the willingness of employees within small to medium sized companies to learn about and integrate AI technologies into their domains. Schwaeye et al. (2024) in *The new normal: The status quo of AI adoption in SMEs* clarified that SMEs face many challenges such as limited financial and personnel resources, that can ease or hinder AI training efforts. Their systematic-literature review of 106 peer-reviewed articles on AI adoption in SMEs, provides valuable insights and clarifies AI implementation within SMEs.

The paper focuses on the point that knowledge and training are important factors in the success of AI integration in the SME space. Employees' positive attitudes and digital proficiency significantly affect the smoothness of AI adoption. When it benefits and aligns with the SME and employee needs and innovation objectives, the likelihood of their enthusiastic adoption increases. Furthermore, a strategic investment in human resources, including AI education and upskilling, is vital for increasing employee willingness to embrace AI. So, looking at how SME employees can be educated on this is important. Overall, these insights point to the direction for a need of AI training that both addresses technical skills and employee concerns about job security and the future of their roles in the world of AI. Schwaeye and Jones (2024)

In *Towards Adoption of Generative AI in Organizational Settings*, Kalyan Prasad Agrawal explores factors influencing the adoption of generative AI in businesses using the Technology- Organization-Environment framework. Six major factors are identified as having a significant impact on adoption: compatibility, complexity, organizational size, regulatory support, environmental uncertainty, and competition intensity. The paper acknowledges the fact that integrating Generative AI into businesses can provide a big advantage, like enhancing decision-making and innovation which could become crucial for the competitive advantage. But doing so presents numerous hurdles, which is why organizations are sometimes hesitant to fully integrate it, they are wary of the complexities and uncertainties surrounding this technology. While larger organizations with more resources are more likely to adopt these technologies. The study highlights that complexity, such as the lack of standards, significantly hinders adoption, while compatibility with existing systems facilitates smoother integration, as it needs to be compatible with existing processes to facilitate the AI integration. Overall, the paper emphasizes that strategic investments in technology infrastructure, human resources, and regulatory alignment are necessary to overcome barriers and successfully adopt generative AI. Agrawal (2023)

## 3 METHODOLOGY

### 3.1 Research Method and Approach

This study will utilize both qualitative and quantitative research methods to assess how employees in small to medium-sized enterprises can be effectively taught about using generative AI. A mixed-methods approach allows for a better understanding of employees' attitudes, knowledge, and preferences regarding usage of generative AI in their roles.

- Qualitative Research: In-depth interviews will be conducted with 5 to 8 SME employees to gather nuanced insights

into their understanding of AI, the challenges they face, and their learning preferences. This will provide context for the quantitative data and highlight individual experiences or concerns.

- **Quantitative Research:** A structured survey, distributed via Qualtrics, will gather data on a larger scale. This survey will target a broad audience of SME employees globally, with the possibility of expanding to include larger enterprises or non-SME participants depending on coordination with other research teams. The survey will include questions designed to assess four key areas:
  1. **Current level of AI knowledge:** Measuring participants' self-assessed knowledge of AI concepts and tools.
  2. **Willingness to learn:** Evaluating how motivated employees are to engage with AI learning opportunities.
  3. **Interest in specific AI topics:** Identifying which areas of AI (e.g., data analysis, machine learning, ethics) employees are most interested in learning about.
  4. **Preferred learning methods:** Understanding employees' preferred formats for acquiring new skills, such as in-person workshops, online courses, or self-paced learning.

### **3.2 Data Collection Methods**

Primary data will be collected using the following methods:

- **Surveys:** The primary tool will be a survey distributed through Qualtrics. Through this survey, quantitative (multiple-choice and scaled responses) data will be collected. The survey will be sent out in September 2024.
- **Interviews:** Our research teams' known people and other, through the process found, SME employees will be invited to participate in follow-up interviews, to gather qualitative data. This allows for deeper exploration of their perspectives on generative AI usage and its relevance to their work.
- **Secondary Data:** Existing literature and research papers on AI training, adoption, and employee development in SMEs will be reviewed to support and contextualize the findings from the primary data.

### **3.3 Sampling Procedures**

Stratified sampling will be used to ensure that diverse employee roles and departments within SMEs are represented in the survey, allowing for analysis across different job functions. A basic random sampling approach will then be applied within each stratum to select participants (or responses) for the survey, ensuring the results are generalizable across SMEs.

### **3.4 Data Management Plan**

The data management plan will outline how survey responses and interview transcripts will be securely stored, processed, and analysed, adhering to data privacy regulations such as GDPR.

### **3.5 Data Analysis Techniques**

- **Descriptive Statistics:** Summary statistics will be used to analyse quantitative survey data, providing insights into overall trends, such as employee interest in generative AI, preferred learning methods, and perceived barriers to AI adoption.
- **Data Visualization:** Graphs, charts, and tables will be used to visually represent findings from the survey, making it easier to compare preferences and attitudes across different employee groups.

### 3.6 Limitations

Potential limitations of the study include the self-reported nature of survey data, which may introduce biases such as social desirability bias. Additionally, given the global distribution of the survey, cultural and regional differences in AI adoption may affect responses. These factors will be considered when interpreting the results.

## 4 PREDICTED OUTCOMES

Four possible outcomes have been predicted for this study. The outcomes will show employees' interest in different AI technologies, their learning preferences in order to effectively teach them about these technologies quickly and on a low budget, what barriers exist when it comes to the subject of AI, and lastly, what is the impact of AI on job performance.

In the first outcome, we expect that the majority of employees (over 70%) will show interest in learning about generative AI. Many of them are likely to recognize its relevance to their current roles and see value in applying AI to their work. AI is becoming more common in business, and employees likely understand that it's important to stay informed about these developments.

In the second outcome, we expect that most employees will likely prefer self-paced learning or online courses to gain AI knowledge, though some may favor in-person workshops or group sessions. We predict that many will express being neutral or slightly uncomfortable with AI tools in their daily tasks. Professionals are increasingly choosing flexible, self-guided learning options. Additionally, unfamiliarity with AI tools may lead to some apprehension.

In the third outcome, we expect that the primary obstacles to learning AI will likely be lack of time and fear of complexity, with fewer employees citing lack of interest or management support as barriers. Employees in small and medium-sized companies often juggle multiple responsibilities, making it difficult to invest time in learning new, complex technologies like AI.

In the last outcome, we predict that most employees will believe that learning AI skills will improve their job performance and career prospects. There will likely be a strong link between employees' confidence in applying AI and their interest in learning more. AI is widely viewed as a way to boost productivity and efficiency, which may drive employees to see it as a valuable career asset.

The findings from this survey will provide valuable insights for designing generative AI training programs tailored to employees in small to medium-sized companies. By addressing common barriers such as time constraints and complexity, and offering learning methods that align with employees' preferences, companies can equip their workforce with essential AI skills, ultimately improving productivity and innovation.

Hypotheses:

1. Employees who are more comfortable with AI tools will show greater interest in learning AI development.
2. Those who identify time constraints as a barrier will prefer self-paced learning options over in-person workshops.
3. Employees who believe that AI will improve their job performance will be more willing to dedicate 3-5 hours or more per week to learning about AI.
4. Employees with less confidence in applying AI concepts will see the complexity of AI as a significant barrier to learning.

## REFERENCES

Agrawal, K. P. (2023). Towards adoption of generative ai in organizational settings. *Journal of Computer Information Systems*.

Schwaeke, J., P. A. K. D. K. K. S. and Jones, P. (2024). The new normal: The status quo of ai adoption in smes. *Journal of Small Business Management*, pages 1–35.

## APPENDIX 1: SURVEY QUESTIONS

### *Section 1: Demographics*

1. How do you describe yourself?
  - ☐ Male
  - ☐ Female
  - ☐ Non-binary / third gender
  - ☐ Prefer to self-describe:
  - ☐ Prefer not to say
2. How old are you?
  - ☐ 10-19
  - ☐ 20-29
  - ☐ 30-39
  - ☐ 40-49
  - ☐ 50-59
  - ☐ 60-69
  - ☐ 70-79
  - ☐ Other:
3. What is your country of residence?
4. How many employees does your company have?
  - ☐ 0-10 employees
  - ☐ 11-50 employees
  - ☐ 50-250 employees

### *Section 2: Current Level of Generative AI knowledge*

1. There are sufficient opportunities within my organization to learn about AI
  - ☐ Strongly disagree
  - ☐ Disagree
  - ☐ Neutral
  - ☐ Agree
  - ☐ Strongly agree
2. How often do you use AI tools or technologies in your work?
  - ☐ Not at all
  - ☐ Less than once a month
  - ☐ Once a month
  - ☐ Several times a month
  - ☐ Once a week
  - ☐ Several times a week
  - ☐ Every day

- ☐ Multiple times per day
3. How familiar are you with the following AI tools on a scale from 1 to 100?
- Chatbots (ChatGPT, Gemini)
- Image generation (DALL-E, Midjourney)
- Speech generation (Descript, ElevenLabs, Resemble.ai)
4. How comfortable are you with the idea of integrating AI tools into your daily work tasks?
- ☐ Very uncomfortable
- ☐ Somewhat uncomfortable
- ☐ Neutral
- ☐ Somewhat comfortable
- ☐ Very comfortable
5. How confident are you in your ability to use generative AI in your current job responsibilities?
- ☐ Not confident
- ☐ Not very confident
- ☐ Neutral
- ☐ Somewhat confident
- ☐ Very confident

### ***Section 3: Willingness to Adopt Generative AI***

1. In your opinion, how important is it for employees in your company to receive formal training in usage of generative
- ☐ Not important at all
- ☐ Not really important
- ☐ Neutral
- ☐ Somewhat important
- ☐ Very important
2. How interested are you in learning about using generative AI in your current role?
- ☐ Not interested at all
- ☐ Not very interested
- ☐ Neutral
- ☐ Somewhat interested
- ☐ Very interested

### ***Section 4: Generative AI Knowledge Interest***

1. What AI topics would you be most interested in learning about?
- ☐ Text generation
- ☐ Image generation
- ☐ Speech generation

**Section 5: Preferences on how to learn about Generative AI**

1. What is your preferred way of learning about the possibilities of AI?
  - ☐ Attending webinars
  - ☐ Attending online courses
  - ☐ Reading articles
  - ☐ Watching videos
  - ☐ Hands-on experimentation (workshop)
  - ☐ Attending conferences or industry events
  - ☐ Listening to audio discussions/podcasts
  - ☐ Engaging in online AI communities
  - ☐ Following AI experts on social media
  - ☐ Not interested in learning about AI
  - ☐ Other (please specify):
2. How much time would you be willing to dedicate to learning about AI per week?
  - ☐ Less than 1 hour
  - ☐ 1 - 2 hours
  - ☐ 2 - 6 hours
  - ☐ More than 6 hours
  - ☐ One-time learning dedication (not weekly recurring)
3. What barriers, if any, do you experience in learning about AI for your role?
  - ☐ Lack of time to learn and practice
  - ☐ Insufficient access to learning resources
  - ☐ Complexity of AI concepts and tools
  - ☐ Lack of support or mentorship
  - ☐ High cost of training programs
  - ☐ Limited understanding of programming
  - ☐ Uncertainty about practical applications in my role
  - ☐ Rapidly changing landscape
  - ☐ Concern about ethical implications of AI
  - ☐ Unsure where to start
  - ☐ Lack of hands-on practice opportunities
  - ☐ No barriers
  - ☐ Other (please specify):